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RESEARCH ARTICLE

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Sensitivity analysis of hybrid microgrids with application to deployed military units

Daniel Reich | **Susan M. Sanchez**

Department of Operations Research, Naval Postgraduate School, Monterey, California, USA

Correspondence
Daniel Reich, Department of Operations Research, Naval Postgraduate School, Monterey, CA, USA.
Email: daniel.reich@nps.edu

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Abstract

We introduce a framework for applying sensitivity analysis to a set of potential hybrid microgrid design options, from which a decision maker can select the most preferred one. In contrast to optimization-centric models that define an objective and produce a single solution, our goal is to empower the decision maker by providing both a range of options and accessible information that enables the decision maker to easily assess their relative upsides and downsides. Moreover, our decoupled approach allows our framework to be paired with any existing model capable of generating a set of potential hybrid microgrid designs. We introduce metrics for computing risk, stemming from power deficits, over a variety of scenarios relating to weather conditions, power demand fluctuations, and extended time horizons. These factors which are inherently uncertain are all important in military operational contexts. By introducing a design of experiments method for sensitivity analysis, we are able to implement parallel processing and maintain computational tractability.

KEYWORDS

design of experiments (DOE), hybrid microgrid, military, sensitivity analysis, simulation

1 | INTRODUCTION

A hybrid microgrid is a collection of distributed renewable and conventional energy resources, such as photovoltaic panels, wind turbines, batteries, and diesel generators. Hybrid microgrids may serve as either short-term or long-term self-sufficient systems for providing energy within a contained geographic area. When integrated into an external energy system, such as a municipal one, a hybrid microgrid may provide supplemental energy that is more environmentally sustainable, more cost effective or both. When operating as the sole source of energy, sizing a microgrid with sufficient capacity is critical for meeting energy demands. The military, in particular, has unique use cases where microgrids must serve as the sole source of energy for deployed units. Our article focuses on these military applications.

[Correction added on 5 Aug, 2023, after first online publication: Handling Editor information added.]

We consider the energy needs of a deployed military unit, operating in varying climate conditions. While an operational mission may be planned for a certain duration of time, we also account for unexpected circumstances that may lead to extended time frames. Such circumstances are key requirements when determining specifications for a deployable microgrid, as a unit stranded without energy could be catastrophic. We also aim to avoid excessive capacity, due to both the associated transportation logistics and cost.

The United States Department of Defense has identified microgrids as a key technology for increasing energy security of the military and for improving environmental sustainability (Van Broekhoven et al., 2013). Microgrid design in the context of military applications has received some attention in the literature (Lucchese et al., 2020). In the context of deployed units, vehicle-based microgrid architectures have been designed for vehicle-to-grid and vehicle-to-vehicle power supply (Abul Masrur et al., 2017), energy management

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systems have been optimized to reduce fuel consumption (Anglani et al., 2017), supercapacitor banks and controller design have been introduced to minimize cost (Oriti et al., 2019), and containerized hybrid energy systems have been analyzed (Berardi et al., 2020). Microgrids have also been explored as a means of improving military base energy resilience (Peterson et al., 2021), including on islands that share some of the self-sufficiency requirements critical to deployed units (Giachetti et al., 2021).

Outside of military applications, the design of general-purpose hybrid microgrids has been an active area of research in recent years (Al-Falahi et al., 2017; Khatib et al., 2016). Particular emphasis has been placed on optimization-focused methods that provide a decision-maker with a single best solution for selecting the sizes of distributed energy resources (DERs) (Alsaidan et al., 2017; Al-Shamma'a & Addoweesh, 2014; Bahmani-Firouzi & Azizipanah-Abarghooee, 2014; Bukar et al., 2019; Cao et al., 2020; Chatterji & Bazilian, 2020; Chen et al., 2011; Dong et al., 2018; Fossati et al., 2015; Hussain et al., 2019; Lai et al., 2019; Lan et al., 2015; Liu et al., 2019; Mashayekh et al., 2017; Rodríguez-Gallegos et al., 2017; Zhao et al., 2014). However, in practice, such methods may fail to capture important but unspecified considerations.

In contrast, when a decision-maker is presented with a range of options for sizing DERs, the process of reviewing those options elicits preferences that may have previously gone unstated when defining system requirements. Methods that provide a range of potential DER sizing solutions allow decision-makers to take a more active role in the process, which in turn may lead to the selection of highly-effective microgrid designs.

Reich and Oriti (2021) introduce a method for generating a set of hybrid microgrid designs with minimally sufficient capacity. In other words, if capacity is reduced from any component of a "rightsized" system, then energy demands would not be fully met at some point in time. While an upside of this rightsized metric is avoiding investment in excess capacity, a downside is that it does not incorporate stochasticity. As a result, the microgrid designs identified may not be robust to changes in weather, energy demands, or other factors.

We extend the work of Reich and Oriti (2021) by proposing a two-stage approach that (i) generates rightsized microgrid designs and (ii) leverages data farming methods (Sanchez & Sanchez, 2017) for sensitivity analysis. This allows us to introduce a flexible notion of robustness, where potential microgrid designs can be tested across an extensive set of different conditions. By doing so, we can identify which designs are less and more robust, and provide this information to a decision-maker. If needed, we can also modify designs with capacity adjustments to increase their robustness.

The goal of our article is to propose a method for designing a hybrid microgrid, given a historical or representative energy demand profile of a facility or operational unit. Although numerous methods for DER sizing have been proposed in the literature, the vast majority focus on reducing the design space to a single solution, in which case the method plays the role of the decision-maker. The method we present is novel in introducing robustness into a decoupled model through sensitivity analysis. The task of balancing energy demands with infrastructure capable of meeting those demands is too computationally challenging for an individual, as is the task of assessing the robustness of a potential microgrid design. On the other hand, a decision-maker is well-suited for comparing potential designs, when presented with metrics of interest, for example, cost, sustainability, risk, and robustness. Our method enables a decision-maker to play a central role in the process, while ensuring the designs considered are viable and well understood.

1.1 | Established software tools

Several well-established software tools have been developed for microgrid planning. HOMER is a proprietary software that performs simulation, optimization, and sensitivity analysis (Lambert et al., 2006). Simulation is used to model the performance of a potential microgrid design, optimization iterates through different potential microgrid designs to identify the one with lowest life-cycle cost, and sensitivity analysis may be performed on almost any input variable that is not a decision variable. HOMER's fundamental capability is simulating long-term operation and it performs simulations for one year on an hourly basis.

In contrast to HOMER's capabilities, our method is ideal for higher-level trade-off studies with objectives other than cost, that is, military mission focused. We place no bounds on either the discretization of time or the intended time horizon, which may be especially useful in military contexts. We perform sensitivity analysis efficiently and easily on a wide range of potential microgrid designs, with the intent of measuring operational effectiveness and risk, which HOMER is not configured to do. Our work is complementary to HOMER's existing capabilities, for example, simulating a detailed electrical analysis for a candidate microgrid design.

DER-CAM produces a single optimized microgrid design, however, it is restricted to a small number of load points, due to the computational tractability of the linear optimization model (Stadler et al., 2014). It also requires all constraints to be linear, which may be a limiting factor for precisely representing power dynamics of distributed energy resources. For example, power output from a battery is a nonlinear function of current and voltage, where current depends on state of charge (Scioletti et al., 2017). While these dynamics can be incorporated directly in a simulation model, a linear optimization model will require the introduction of simplifying assumptions.

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XENDEE is proprietary software that, like DER-CAM, employs linear optimization techniques and identifies a single solution (Stadler & Naslé, 2019). XENDEE's and DER-CAM's optimization techniques do not allow for detailed sensitivity analysis, for example, to understand the impact of uncertainty in weather conditions. Like HOMER, XENDEE includes capabilities for detailed electrical analyses, which we see as complementary to our work when used after one or a small number of potential microgrid designs of interest have already been selected.

Microgrid Design Toolkit offers two primary capabilities (Arguello et al., 2015). The first is based on the DER-CAM model and uses linear optimization to provide a single solution for microgrid size and composition. The second is intended to refine the design by (a) applying a genetic algorithm to generate a set of candidate designs and (b) applying a simulation-based reliability model to assess the performance of those designs. Our work employs a similar simulation strategy and is complementary to their analyses by allowing factors to be incorporated that are not considered in their performance and reliability model, for example, the impact of weather uncertainty on performance.

REopt (Renewable Energy Integration and Optimization) also uses an optimization approach to identify hybrid microgrids that minimize expected costs or energy resilience for an installation with a specified load profile (Anderson et al., 2017). It defines resilience as "the amount of time on-site generation and storage can sustain the site's critical load during a grid outage," where the grid outage is predictable so all the batteries are fully charged when the outage begins. The optimization assumes that solar irradiation, wind speed, and both electrical and thermal load profiles are perfectly predictable, and fails to take into account some operational considerations, such as short cycling for boilers or differences among PV technologies. REopt has a simulation feature to further explore cost, energy usage, and resilience over a 1-year period. We use a similar characterization of resilience by investigating the length of time beyond the nominal duration for which operations can proceed with high probability.

1.2 | Novel contribution

When designing a microgrid, planning may be executed in stages. At an early stage, the goal may be to identify the sizing of various distributed energy resources. At a later stage, once a high-level design or small set of candidate high-level designs are selected, both the placement of these resources and the detailed electrical schematics required may be defined. Further analyses may be conducted, focusing on reliability, resilience, and other important considerations.

Our framework is intended for the earlier planning stages, where higher-level trade-off studies may allow decision makers to assess a wide range of potential microgrid design options and narrow them down to a single preferred solution or a small set of candidate solutions. To facilitate an iterative design process, where a decision maker can play a central role in considering trade-offs, we develop a decoupled framework for sensitivity analysis of microgrid designs. By decoupling, we allow capabilities of other methods and tools to be utilized in complement with our work, for example, methods for identifying a set of potential hybrid microgrid designs; the detailed electrical analysis capabilities offered by HOMER and XENDEE; and the performance and reliability model in the Microgrid Design Toolkit or other resilience and reliability models (Giachetti et al., 2021; Mishra et al., 2020; Peterson et al., 2021).

Our focus is on a specific military context where a microgrid is being designed for short-term deployment. Unlike more common use cases, cost is not the primary consideration. Common concerns for reliability or resilience may be different in this context, for example, at a base one may consider the effect of natural disasters on infrastructure that may not be relevant in a short-term military deployment. Similarly, reliability of equipment over a two-week time horizon may be unique to our military application context. Existing tools and methods, which have been designed for more standard use-cases of planning stationary infrastructure intended for long-term use, may not be ideal or incorporate all important considerations for deployed military unit microgrid planning.

A key benefit of our approach is the use of data farming as a highly efficient, structured means of sensitivity analysis. To the best of our knowledge, this approach is new to those involved in hybrid microgrid design. Previous approaches involve applying optimization to a single location to provide a single solution (e.g., Zhao et al., 2014), comparing a solution to a current base case to estimate average cost savings (e.g., Anderson et al., 2017), making comparisons among a small set of potential alternatives (e.g., Hassan et al., 2018; Mashayekh et al., 2017; Peterson et al., 2021), or conducting a series of small comparative experiments where only one or two factors are varied at a time (e.g., Anderson, 2020; Giachetti et al., 2021). Our approach is sequential. First, we select potential microgrids. Next, we subject these to sensitivity analysis to assess their performance in the presence of a wide range of uncertainties. Finally, we identify a subset of those high-performing candidate microgrids as potentially desirable solutions. The aforementioned approaches are all fundamentally different and can lead to very different microgrid recommendations.

When applied in conjunction with existing complementary tools and methods, our framework is sufficiently general that it may provide value when planning both specialized and more standard use cases. Using simulation provides modeling flexibility, including nonlinear power dynamics for DERs and complex control logic for energy management systems.

The remainder of this article is organized as follows. In Section 2, we summarize the method we employ for identifying potential microgrid design options. In Section 3, we introduce variability and define a framework for measuring

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deficits and risk. In Section 4, we describe our sensitivity analysis method for assessing the performance of potential microgrid designs. In Section 5, we summarize our framework. In Section 6, we demonstrate our method through computational experiments. Our conclusions appear in Section 7.

2 | IDENTIFYING POTENTIAL MICROGRID DESIGNS

We consider hybrid microgrid designs consisting of three DERs: a solar photovoltaic system (PV), a battery energy storage system (BESS), and diesel generation (DG). The central decision in the planning process is the level of capacity to select for each DER. Given a baseline energy demand profile, many potential microgrid designs would be capable of meeting the energy needs. For example, one design may be reliant on solar photovoltaic power during the day and battery power at night, whereas another may be reliant solely on diesel generation. Each potential design has advantages and disadvantages, which can be summarized using different metrics. For example, designs with higher photovoltaic and lower diesel generation capacities may be advantageous from an emissions standpoint, but disadvantageous in terms of installation cost. Higher battery capacities may also be advantageous in terms of emissions, but batteries may degrade at a faster pace than diesel generators.

Our first step is to identify the range of potential designs, which we accomplish using the rightsizing method of Reich and Oriti (2021), summarized in Figure 1. This method is an iterative approach that uses nested binary search algorithms to adjust DER capacities until two conditions are met:

1. Total microgrid capacity is sufficient to meet demand at all points in time.
2. A reduction in any of the DER source capacities would yield insufficient total microgrid capacity to meet power demand at some point in time.

The outer loop of the rightsizing method is the simplest. It iterates over the level of DG capacity, starting with the specified minimum and increasing by a specified step size, until capacity is sufficient to meet peak load with no other DERs.

For a given level of DG capacity, the inner loops of the algorithm use binary search to identify combinations of PV and BESS capacities that meet the two conditions stated above. BESS capacity is initialized to 0 and PV capacity is initialized to infinity. If a power deficit exists at some point in time with infinite PV capacity, then more BESS capacity is required and is added. When power demands are met at all points in time, PV capacity is decreased to remove excess capacity and the design obtained is added to the set of potential microgrid designs.

At the next iteration of the inner loop, BESS capacity is increased, ensuring PV capacity need not increase. In other words, if PV capacity was sufficient at the prior iteration with less BESS capacity, then it will remain sufficient with more BESS capacity. If PV capacity cannot be decreased without a power deficit arising at some point in time, the iteration of the inner loop produces a dominated design, so BESS capacity must be increased again. Otherwise, PV capacity decreases and another potential microgrid design is identified. This process repeats until PV capacity is as close to 0 as the specified step size allows, without PV capacity becoming negative, at which point the outer loop increases DG capacity to the next level.

Testing whether a power deficit exists at any iteration of the algorithm is accomplished by simulating the operation of a microgrid with the DER capacities associated with that iteration. In this context, the simulation is deterministic: a single baseline power demand is used throughout the process. In our computational experiments, we employ a simple energy management system with four states used in priority order: (1) photovoltaic power while BESS charges; (2) photovoltaic and BESS power; (3) photovoltaic and diesel power while BESS charges; (4) photovoltaic, diesel, and BESS power. An upside

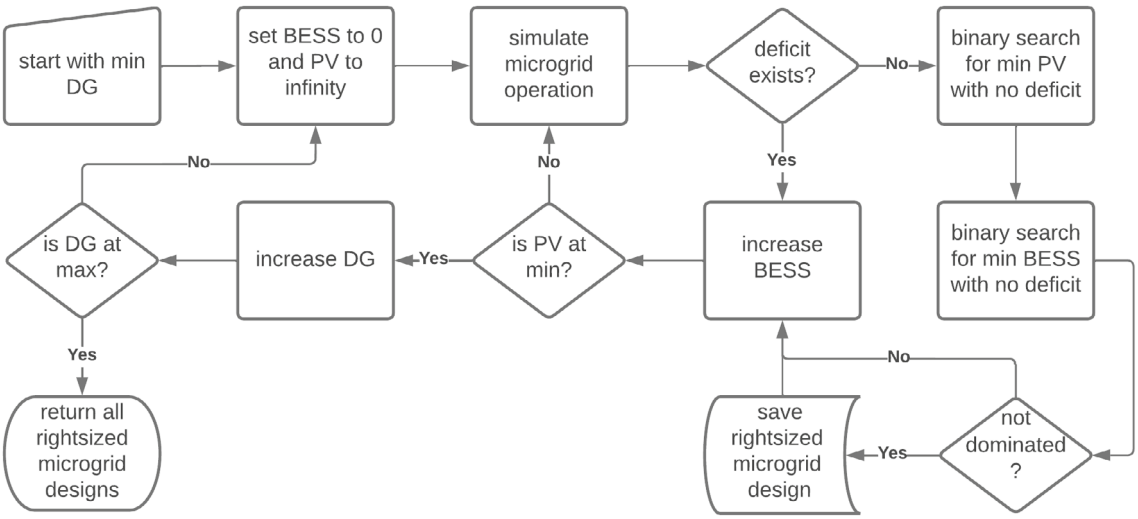

FIGURE 1 Rightsizing method of Reich and Oriti (2021) to identify hybrid microgrid designs capable of satisfying a given energy demand profile.

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of our methodology is that it places no limitations on the energy management system, so alternative operational controls can be introduced. For further detail on the simulation model, please refer to Reich and Oriti (2021).

3 | VARIABILITY, DEFICITS, AND RISK

Potential microgrid designs produced using the rightsizing model described in Section 2 all satisfy the energy demands of the baseline scenario used. However, their performance under varying conditions may differentiate one potential design from another. For example, if power demands are higher than expected during the night, designs with large DG capacity may have the excess capacity available during those nighttime hours to satisfy those higher demands. If weather conditions are generally more cloudy than those represented in the baseline scenario, designs with large PV capacity will have degraded performance. If the duration is longer than anticipated, then designs with small PV capacity (i.e., lower renewable power generation capacity) will have degraded performance.

We examine three different sources of random variation: uncertain demand, uncertain weather, and uncertainty in the operational duration. We first describe how we parameterize these random components to provide flexibility as we systematically explore their impacts. We then define sensitivity and risk in this context.

3.1 | Varying power demands

Let $p_l(\cdot)$ denote the power load demand for the baseline scenario used to identify the microgrid designs in Section 2. We randomly generate a scenario for power load demand corresponding to a random variable Y, such that

$$Y \sim \max\{0, \mathcal{N}(p_l(t), \delta p_l(t))\} \tag{1}$$

and $0 \leq \delta \ll 1$ sets the standard deviation $\sigma = \delta p_i(t)$ of the Normal distribution to a small fraction of its mean. Conceptually, this corresponds to situations where our power load demands are nearly correct, but small fluctuations might occur in practice.

3.2 | Varying weather

Reich and Oriti (2021) model the PV power available at time t using the positive part of a sine-wave equation:

$$p_{pv}(t) = \max\left\{0, \frac{q_{pv}\alpha_t}{2w}\left[2w - 1 - \sin\left(\frac{\pi(t+6)}{12}\right)\right]\right\}, \tag{2}$$

where q_{pv} is the PV power rating that sets the base wave amplitude, t is the time in hours, α_t is a scaling percentage based on weather conditions at time t, and w adjusts the length of the daylight period based on location and season ($0 < w < 0.5$ has shorter days than nights, while $0.5 < w < 1$ has longer nights). In their experiments, they fix $\alpha_t = \alpha, \forall t$, focusing on rightsizing a grid for operations in a uniformly sunny setting.

We introduce a PV model with two random weather components: g_t controls for daily fluctuations in weather conditions and ξ_t controls fluctuations at each point in the time discretization, a finer time scale. This two-component construction has the ability to capture sunny days versus cloudy days versus rainy days, as well as the more subtle changes in weather that may occur throughout any day. The discretization converts between continuous time t to the time step (Δ) (in fractions of hours), but for ease of notation we do not explicitly include a time step index in the equations that follow. For the daily variation,

$$g_t = u_{\lfloor t/24 \rfloor}, \tag{3}$$

where $b \in [0,1]$, $c \in [0,1]$, and u_i is generated from a $\mathcal{U}(cb, b)$ distribution for day i, $i \in \{0, 1, 2, \ldots\}$. If $b = c = 1$ then the day is completely sunny. If $b = c = 0$ the day is completely cloudy and PV cells will be ineffective. Using other values of b and c will control the minimum and maximum potential cloud cover for a day within the specified operational window.

At the finer time scale, we apply an autoregressive AR(1) model of the form:

$$\xi_t = \phi\xi_{t-\Delta} + (1-\phi)x_t, \tag{4}$$

where ξ_t determines short-term weather variations at time t (in hours), Δ is the length of the time step, $\phi \in [0,1]$ is the autoregressive coefficient, and x_t is a solar scaling weight at time t generated from random variable $X \sim \mathcal{N}(\mu, \sigma)$. Note that this model can be used to test represent unpredictable weather patterns in areas where they may occur, by setting the autoregressive parameter to a low value, $\phi \ll 1$. AR models are special cases of autoregressive integrated moving average (ARIMA) models that have been used to represent short-term cloud cover patterns (Wang & Wang, 2018).

Replacing α_t from (2) with the product of g_t and ξ_t from (3) and (4), respectively, the PV power available at time t becomes:

$$p_{pv}(t) = \max\left\{0, \frac{q_{pv}g_t\xi_t}{2w}\left[2w - 1 - \sin\left(\frac{\pi(t+6)}{12}\right)\right]\right\}. \tag{5}$$

Here, q_{pv} is the PV power rating that sets the base wave amplitude, t is the time of day in hours, g_t adjusts the magnitude of the wave for an overall cloudiness or sunniness level for day $\lfloor t/24 \rfloor$, and w ($0 < w < 1$) adjusts the length of the daylight period based on location and season as before. In other words α_t from (2) is equal to $g_t\xi_t$ in (5). One final note: the use of $\sin(\pi(t+6)/12)$ in (2) sets the phase shift to correspond to peak solar insolation at midday. More generally, this is $\sin(\pi(t + peak/2)/12)$ where $peak$ is the hour of peak insolation for a 24 h day. For illustrative purposes, we fix the peak time at noon ($peak = 12$) throughout this article.

Examples of PV power curves from (5) appear in Figure 2 for a 4-day period; these incorporate variability in cloud

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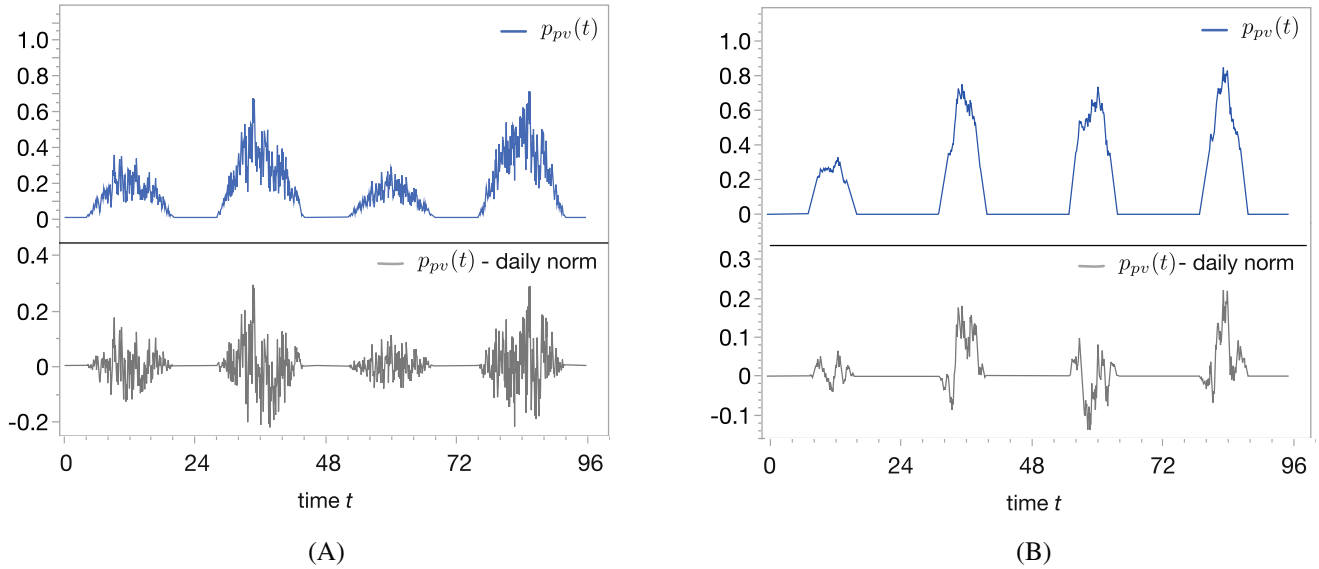

FIGURE 2 Examples of simulated weather patterns, where $p_{pv}(t)$ denotes photovoltaic power (in kW) at time t (in hours). Short-term weather variations are generated every four minutes. (A) Long days with moderate autocorrelation ($\phi = 0.5$). Short days with high autocorrelation ($\phi = 0.95$).

cover from (3) and (4). With different choices of parameters, these can produce synthetic cloud cover data similar to the variety of fine-scale patterns of collected data (see, e.g., Jain, 2020).

We sample from normal and uniform distributions for the power load demand and weather scaling factors because these distributions are intuitively simple when data is not available to inform precise distributional assumptions. However, our sensitivity analysis method does not rely on the mathematical properties of these distributional assumptions. Other distributions can be employed without requiring further modifications to our sensitivity analysis method. Initially, we want to see what matters and what difference it makes to include these sources of variation.

3.3 | Varying operational duration

Although the operational duration is considered fixed in the earlier discussion, in reality there might be some uncertainty associated with how long the unit will be deployed. Durations that are longer than anticipated will have higher total power demands, as well as more opportunities for the randomly generated demands or weather conditions to take on unusually high or low values. A potential microgrid that has been designed for a 14-day period, but does not operate effectively for at least several more days under similar conditions, would be an unsatisfactory solution for military deployments. By subjecting all microgrids under consideration to extended simulation, we can assess the robustness of their performance in this regard.

Notationally, let D_0 denote the nominal operational duration but suppose there are alternative durations of interest. In our computational experiments in Section 6, we use a nominal duration of 2 weeks (D_0) and a single alternative duration of 3 weeks (D_1).

3.4 | Defining deficits and risk

Incorporating uncertainty in demand, weather, and duration means that there is no longer a deterministic solution to the microgrid selection problem. Instead, we would ideally assess whether or not the following inequality is satisfied:

$$\text{Prob}(Z_j \leq z^*) \geq 1 - \epsilon, \tag{6}$$

where Z_j is a random variable representing the proportion of time during an operation that the microgrid design $j \in S$ fails to meet demand (i.e., the deficit), $z^* : 0 \leq z^* \ll 1$ is the highest acceptable probability of deficit during a single operation, and $\epsilon : 0 \leq \epsilon \ll 1$ is the highest acceptable risk tolerance. If $z^* = 0$, demand must be satisfied throughout the operation with probability $1-\epsilon$. Restated in terms of standard reliability indices for power grids (see, e.g., NERC, 2018), by setting $z^* = 0$, the left-hand side of Equation (6) measures the loss of load probability (LOLP) and the right-hand side provides an acceptable threshold for LOLP. If $\epsilon = 0$, the deficit proportion must be always be z^* or lower. It is important to note that both the acceptable deficit proportion z^* and acceptable risk threshold ϵ are thresholds chosen by the decision maker, and may vary widely from one application context to another. For example, if power must be provided to keep heart monitors running in a mobile medical unit, then both ϵ and z^* should be small. If a certain level of deficits can be tolerated, but larger deficit probabilities must be avoided, then a larger z^* with a small ϵ might be more appropriate; this might correspond to accepting that A/C units are sometimes able to cool rooms only to a few degrees above the desired temperature level, but there is little risk of a complete power outage. Alternatively, a decision maker willing to halt or postpone an operation if conditions are unfavorable might use $z^* = 0$ with a larger ϵ to assess whether or not the operation is likely to go off without any hitches attributable to lack of power.

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Let ω denote a scenario consisting of the operational duration D^ω, the power load demands $\{p_l^\omega\}$, and weather conditions $\{\xi^\omega,\ g^\omega\}$. Let $z_{j,i}^\omega$ denote the deficit for microgrid design j and replication i in scenario ω ($z_{j,i}^\omega \geq 0$). Then for any user-defined threshold z^*, we can estimate the probability (risk) the power deficit is unacceptably large using microgrid j in scenario ω as:

$$Risk_{j,z^*}^\omega = \frac{1}{n_r}\sum_{i=1}^{n_r}\mathbb{1}\left(z_{j,i}^\omega > z^*\right) = 1 - \frac{1}{n_r}\sum_{i=1}^{n_r}\mathbb{1}\left(z_{j,i}^\omega \leq z^*\right), \quad (7)$$

where $\mathbb{1}(\cdot)$ is the indicator function and n_r is the number of replications. The superscripts on $Risk^\omega$ and z^ω indicates these are for a particular scenario. If we simulate this for multiple scenarios $\omega \in \Omega$, our estimate of risk for microgrid j, given a specified z^*, is the sample average

$$Risk_{j,z^*} = \frac{1}{|\Omega|}\sum_{\omega\in\Omega} Risk_{j,z^*}^\omega. \quad (8)$$

Microgrid j remains a viable option if $Risk_{j,z^*} \leq \epsilon$ for the desired z^* and ϵ. When running simulations, one can output the deficits z for all microgrid designs in all replications and scenarios. Accordingly, these risks can be recomputed for different combinations of z^* and ϵ, by conditionally summing those replications where a desired deficit level of interest is not met, without additional simulation.

Although resilience is an important consideration for power grids, the development of resilience metrics is ongoing (Mishra et al., 2020). Our notion of risk is related to resilience as described in Anderson et al. (2017). They quantify resilience in terms of the average length of time (in days) the grid is able to satisfy a specified probability of surviving an outage ($1 - \epsilon$ in our notation) and so can assess a "resilience benefit" of one hybrid system over another in terms of additional average operational days. Rather than fixing the probability and calculating the average additional days, we fix one or more durations and calculate the probabilities of remaining operationally viable. For those who define resilience as the time or average time between outage and recovery, we note that setting $z^* > 0$ allows for one or more short outages during an operation while limiting the total downtime.

4 | ASSESSMENT METHODOLOGY

We now describe a sensitivity analysis method to systematically test the performance of potential microgrid designs in the presence of uncertainty. We use a design of experiments (DOE) framework suitable for large-scale simulation studies, an approach called *data farming* since we "grow" data from the computational models is an efficient, structured manner. We refer the interested reader to Sanchez and Sanchez (2017) or Sanchez et al. (2021) for more information on data farming; see (Kleijnen, 2015; Law, 2015), or Nelson (2013) for other discussion of DOE for simulation studies. The website of the SEED Center for Data Farming (2020) has links to many other data farming resources and examples. In DOE parlance, each input to our simulation model that we choose to vary is called a *factor*. A factor can be a model input, such as the autoregressive coefficient ϕ in (4). Alternatively, it can describe the relationship between two model inputs, such as the δ that modifies the standard deviation in (1). Factors can be continuous-valued (e.g., ϕ), discrete-valued (e.g., an integer number of days), or qualitative (e.g., the set of microgrids under investigation).

The ability to effectively assess the sensitivity and risk of a rightsized microgrid design identified through our approach in Section 2 is influenced by our selection of parameters for the distributions governing power load and weather in Sections 3.2–3.4, as well as the duration of operations. While it is possible in some application contexts that historical data may be available to identify reasonably precise power load and weather distributions, in most application contexts this may not be possible. For example, consider a military unit being deployed to an area where no prior similar missions have been conducted and with unpredictable weather patterns.

When precise distributions are unidentifiable, a method to assess sensitivity of microgrid design performance to sensitivity of distribution parameter selection provides useful information. A straightforward way to layer such an approach might be to identify all potentially relevant distribution parameters and test all possible combinations, where assessing sensitivity of microgrid design performance becomes a subproblem for each set of distribution parameters. However, this type of combinatorial approach (also called a full factorial design, or gridded design) is only possible when we bound the number of potential settings for each parameter—and even so, it quickly becomes computationally intractable as the number of factors increases (Sanchez et al., 2021). Let $\ell(\phi)$ denote the number of levels of interest for the autoregressive coefficient ϕ in (4), and similarly for the other parameters. Then the total number of combinations (each called a design point and abbreviated dp) a brute force approach requires can be found by multiplying the number of potential levels for each factor as follows:

$$\begin{aligned}\#dp = {} & \#(\text{microgrids},\ S) \times \#(\text{locations},\ w) \\ & \times \#(\text{start dates}) \times \#(\text{std dev multipliers},\ \delta) \\ & \times \#(\text{cloud upper bound},\ c) \times \#(\text{cloud multiplier},\ b) \\ & \times \#(\text{solar scaling mean},\ \mu) \\ & \times \#(\text{solar scaling std dev},\ \sigma) \\ & \times \#(\text{autoregressive coefficient},\ \phi) \\ & \times \#(\text{durations},\ D),\end{aligned}$$

or written more succinctly as

$$\begin{aligned}\#dp = {} & (\#S) \times (\#w) \times (\#peak) \times (\#\delta) \times (\#c) \\ & \times (\#b) \times (\#\mu) \times (\#\sigma) \times (\#\phi) \times (\#D). \quad (9)\end{aligned}$$

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For stochastic simulations, the total number of runs is the product of the number of design points and the number of replications.

The computational requirements are radically reduced when making use of efficient experiment designs, bringing these studies from the realm of the impossible to the realm of the practical. This will be clear in our computational example. The most efficient designs contain no crossing, but consist of carefully chosen dps within the multidimensional space; this is sometimes called a combined design (Kleijnen et al., 2005; Myers et al., 1992). However, constructing the overall design by crossing two or three smaller designs has some advantages for visualization and interpretation, because it facilitates benchmarking. It may be particularly useful when seeking robust or resilient solutions when separate designs are used for controllable factors such as the rightsized microgrid, and uncontrollable factors such as weather-related distributional parameters (Sanchez & Sanchez, 2020).

For our study, a smaller overall design could be constructed by crossing a design for the microgrids with a single design for the other nine factors, as shown in Equation (10).

$$\#dp = (\ell_S) \times \underbrace{(\ell_w),\ (\ell_{peak}),\ (\ell_\delta),\ (\ell_c),\ (\ell_b),\ (\ell_\mu),\ (\ell_\sigma),\ (\ell_\phi),\ (\ell_D)}_{\text{efficient DOE for other factors}}. \tag{10}$$

In our model, the power demands and weather conditions are simulated independently. This might not always be appropriate, but we separated it for simplicity. However, if the connection between the two were important for a particular operation (e.g., power needed for A/C is more on sunny days), that could be taken into consideration with minor changes to the model code.

5 | PROCESS FLOW

Figure 3 summarizes our sensitivity analysis framework for generating and assessing potential microgrid designs options. The process begins with a target power load, which is the baseline demand for any potentially viable microgrid design option. Next, a method for identifying potential microgrid designs is executed. Here, we use the method of Reich and Oriti (2021) as described in Section 2, but any other suitable method may be employed. Optionally (not included in Figure 3), additional potential microgrid designs may be constructed by increasing capacities of the designs identified. The resulting set of alternatives are then subjected to the sensitivity analysis method we propose in Section 4, using an input set of design points for running the design of experiments. The deficit and risk measures introduced in Section 3 are computed and then analyzed, as presented next in Section 6. Low-performing potential designs are eliminated. A decision-maker reviews the remaining set of microgrid design options and provides input on which ones to prioritize. These selected designs can then be subjected to further analysis using complementary planning tools, as discussed in Section 1.

6 | COMPUTATIONAL EXAMPLE

6.1 | Baseline settings for identifying rightsized microgrids

Although the method we propose can be applied to any standalone microgrid, our computational experiments are motivated by a military deployment with a planned duration of two weeks. We use the baseline power load profile from Reich and Oriti (2021) shown in Figure 4, which follows our convention of negative power values representing loads and battery charging, to differentiate from positive-valued power generation and battery discharging. The time horizon is discretized into 4-min intervals, which translates to 5040 time steps at which microgrid operation needs to be simulated.

However, in the context of a military mission, unexpected circumstances may lead to extended deployments and microgrid designs that are robust to accommodate such changes may be required. So, when applying our sensitivity analysis method from Section 4, we extend the time horizon to 3 weeks or 7560 time steps of 4-min duration.

We set input step sizes of 20 kW for diesel generation and photovoltaic power, and 20 kWh for the battery energy storage system. We initialize the battery energy storage system to a fully charged state $\gamma = \overline{\gamma} = 1$ when simulating microgrid operation and set the minimum allowable state of charge $\underline{\gamma} = 0.2$. The rightsizing method produced the 65 potential microgrid designs shown in Figure 6. Note: we use a larger step size than was used in Reich and Oriti (2021), so the number of potential microgrid designs identified is reduced from 1283 to 65. By setting the weather distributions in our baseline scenario to a fixed point value, our parameters otherwise match those of Reich and Oriti (2021). A plot of the power supplied or stored over time for the rightsized microgrid design with 60 kW diesel generation capacity, 260 kWh BESS capacity and 120 kW photovoltaic capacity is shown in Figure 5.

Figure 6A shows a scatterplot of the BESS energy versus PV power, where the marker shapes and colors indicate the diesel generator capacity. We plot using square-root scaling on the axes to make it easier to differentiate among the less-extreme solutions. In Figure 6B, the rightsized microgrids are (from top to bottom) in increasing order of BESS, and then increasing order of diesel. The largest bar for each component corresponds to the maximum value that appears in the 65 rightsized designs. The scaling is component-wise, with the maximum for all three DER types receiving the same width, which represents 100%. For example, the top bar contains the maximum photovoltaic capacity (3200 kW)

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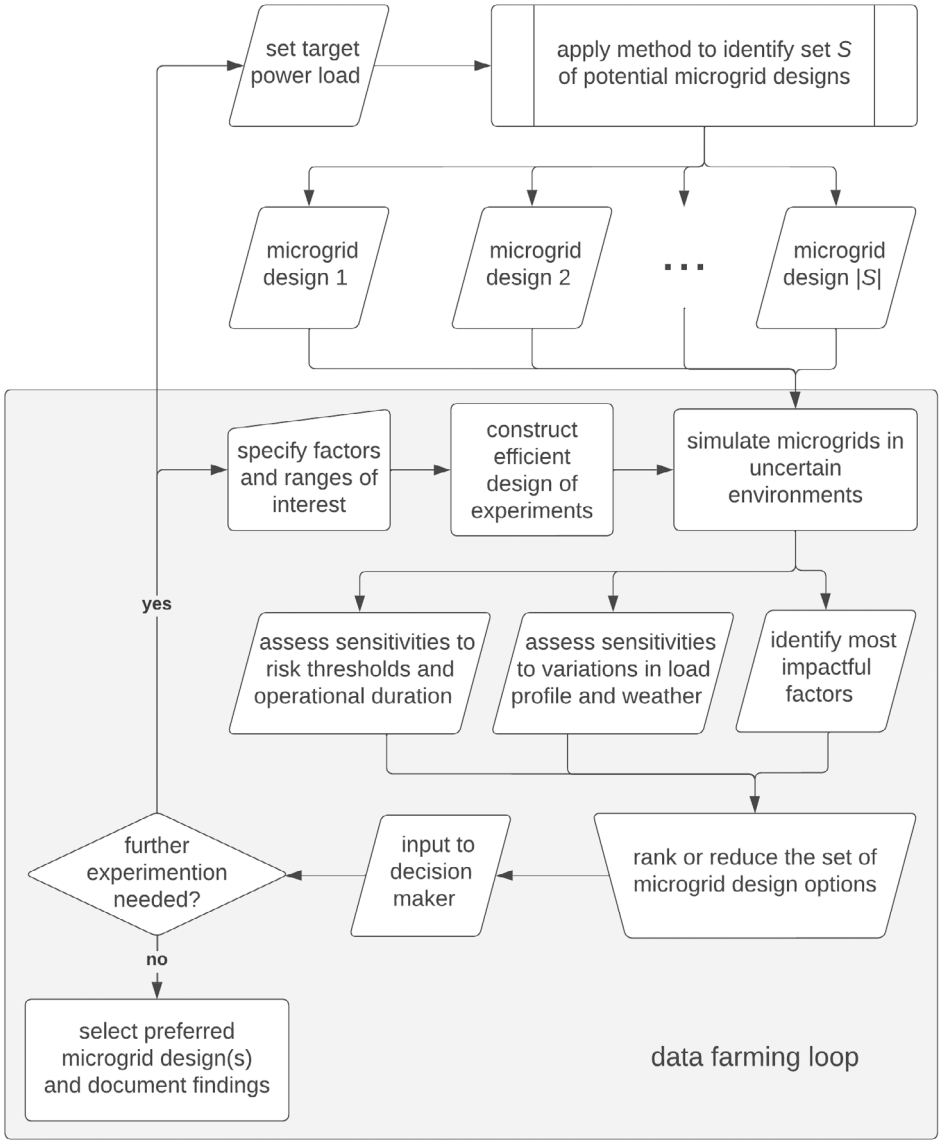

FIGURE 3 Process flow diagram for our sensitivity analysis framework.

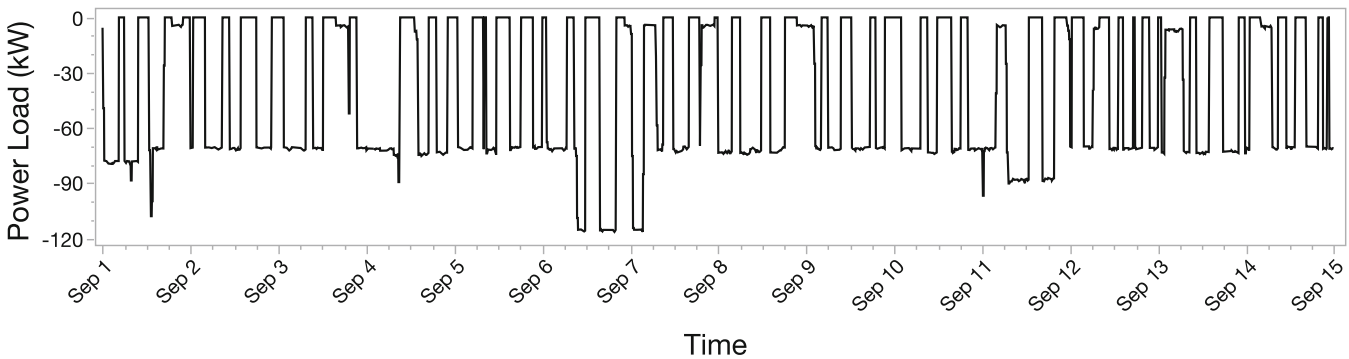

FIGURE 4 Power load profile over two-week time horizon for baseline scenario from Reich and Oriti (2021).

on the right in blue (100% of the maximum photovoltaic) plus a small amount of BESS capacity (860 kWh) on the left in red (~5% of the maximum BESS), so the total width of the top bar (~105%) exceeds the total width of the pure BESS (17 480 kWh) and pure diesel (120 kW) microgrids, which measure 100% of the maximum for their respective types.

6.2 | Experiment setup

Picking up on (9), we cross three smaller designs instead of crossing the levels for each factor. For illustrative purposes, we fix the location, start date, and baseline power demand profile so $\ell_w = \ell_{peak} = \ell_p = 1$. We use ℓ_D=2, gathering the sensitivity and risk results after two weeks and three weeks of operation.

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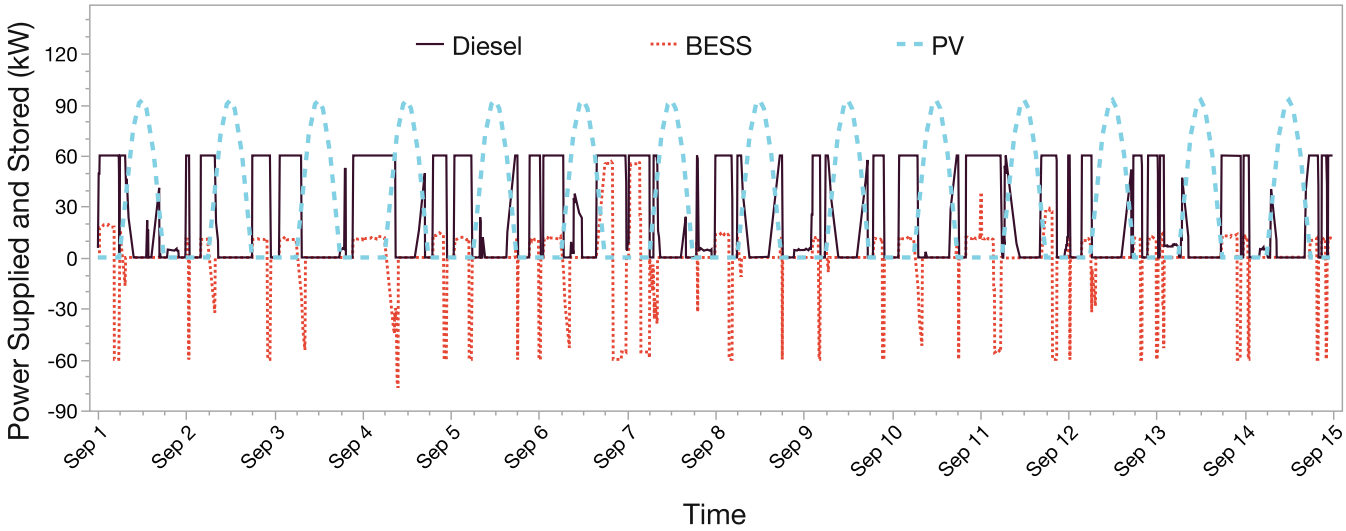

FIGURE 5 Power supplied or stored over two-week time horizon for the baseline scenario from Reich and Oriti (2021) with a microgrid design of 60 kW diesel generation capacity, 260 kWh BESS capacity, and 120 kW photovoltaic capacity.

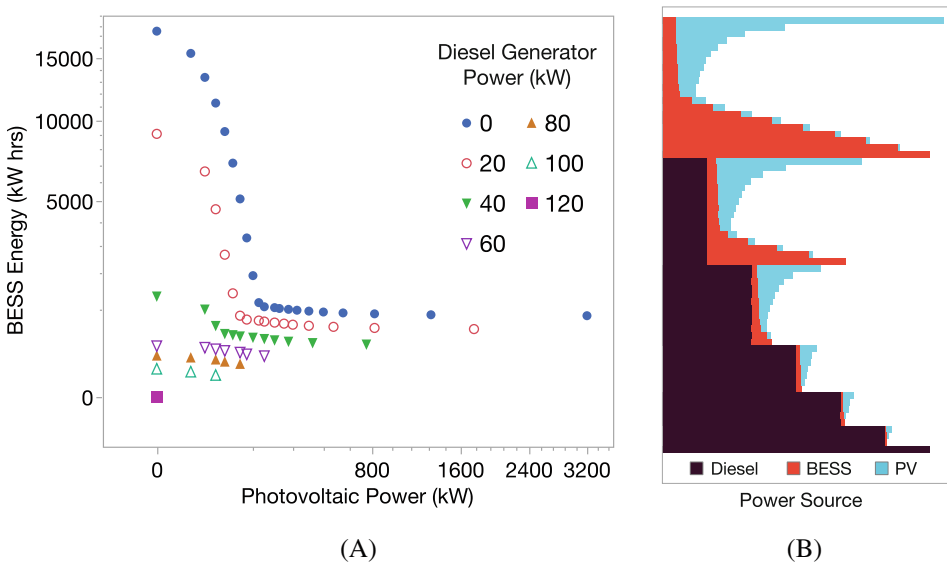

FIGURE 6 Visual representations of microgrid designs rightsized to meet baseline power load requirements. (A) Microgrid composition, plotted on square root axes to provide differentiation. (B) Microgrids with component-wise scaling proportional to the maximum among the 65 rightsized designs.

$$\#dp = \underbrace{(\ell_S),\ (\ell_w),\ (\ell_{peak}),\ (\ell_p)}_{\text{\# baseline rightsized microgrids}}$$

$$\times \underbrace{(\ell_\delta),\ (\ell_c),\ (\ell_b),\ (\ell_\mu),\ (\ell_\sigma),\ (\ell_\phi)}_{\text{efficient DOE for distributional parameters}} \times \underbrace{(\ell_D)}_{\text{\# durations}}$$

$$= 65 \times 257 \times 2 = 33\ 410. \tag{11}$$

The DOE we used for the distributional parameters is comprised of a single stack of a nearly orthogonal Latin hypercube design capable of exploring up to 29 factors at 257 levels each (an NOLH(29 257)) generated by publicly available software (Sanchez, 2021). Other designs could have been used, such as one to seven stacks of an NOLH(7,17) or one to 100 stacks of an NOLH(100 512). For comparison purposes, a brute force approach for five factors at 257 levels each would require 1.12×10^{12} dps or over 7 orders of magnitude more sampling. Even reducing the number of levels to 17 for each factor requires 1 419 857 dps for a brute force approach, or over 4 orders of magnitude more sampling than we actually used. We did not choose the NOLH(7,17) because the NOLH(29 257) is quite efficient and we preferred the analysis flexibility it provides.

We generate independent identically distributed random power loads according to the distribution in (1), where $\delta = 0.1$ so that the standard deviation at any point in time is 10% of the power load at that point in the baseline scenario in Figure 4. For the extended time horizon, that is, $t > 336$ h, we assume $p_l(t) = p_l(t - 336)$; in other words, the baseline power load in the third week of the extended baseline scenario is assumed to be the same as the first week. However, the random perturbations are different.

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TABLE 1 Range of weather distributional parameters considered in computational experiments.

	Daily cloud distribution		Solar scaling distribution		Autoregressive coefficient
	$U \sim \mathcal{U}(a = cb, b)$		$X \sim \mathcal{N}(\mu, \sigma)$		
	c	b	μ	σ	ϕ
Lower bound	0	0	0.3	0.05	0.7
Upper bound	1	1	0.7	0.15	0.98

When applying the rightsizing method from Section 2 to determine the initial set of potential microgrid designs, we fix the weather conditions with $g_t = \xi_t = 0.5$. When applying our sensitivity analysis method from Section 3, we consider unpredictable weather patterns with a range of distribution parameters shown in Table 1. The daily random variables for cloud cover in (2) are uniformly distributed, with the upper parameter of the distribution varied between 0 and 1, and the lower parameter as a fraction of the upper parameter. The random variables for sun intensity scaling in (4) are normally distributed with mean varying from 0.3 to 0.7 and standard deviation varying from 0.05 to 0.15. The autoregressive coefficient in (4) varies from 0.7 to 0.98.

We run a design of experiments to generate 257 distribution parameter combinations. For each parameter combination, we construct 50 new scenarios, by generating 50 weather samples and 50 corresponding power loads. The resulting scenario set contains 12 850 instances on which we simulate microgrid operation to measure performance. Each of the 65 microgrid designs is analyzed, yielding a total of 835 250 microgrid simulations, with each simulation covering the 3-week time horizon.

Additionally, we consider microgrid design modifications, in which capacity of initial designs is increased by 5%, 10%, 15%, or 20%, with the capacity increases allocated to the photovoltaic system and/or the battery energy storage system in these same increments. The resulting 11 435 900 microgrid simulation instances illustrate the need for the DOE approach we employ for computational tractability.

6.3 | Implementation and run time

Given the scale of our computational experiments, parallel processing is needed to solve all instances in a reasonable time frame. Our methods are implemented in Python and the analysis of a single microgrid requires approximately 6 hou of run time to complete all 12 850 subproblem instances. We use around 190 compute nodes on our institution's high performance computing center; however these nodes may have different CPU specifications. All computations are completed in under 35 h of clock time or 6650 h of total compute time. Without the roughly five orders of magnitude saved by using a DOE approach, the clock time alone would have been over 17 million years or 22 years if we had attempted brute-force exploration of the five weather factors at 257 or 17 levels, respectively.

After an initial look at our data, we opted to retain and analyze only those runs corresponding to an upper limit on the daily cloud cover factor $b \geq 0.3$. Our resulting subdesign for the distributional factors has 180 dps (instead of 257) and is still nearly orthogonal, with a maximum absolute pairwise correlation of 0.0469. The flexibility to focus on subsets of the space is one benefit of using a large space-filling design for the initial runs.

6.4 | Measuring sensitivity of microgrid designs

We now illustrate several ways to assess the sensitivity of the baseline rightsized microgrid designs to various types of uncertainty, and various risk preferences of the decision maker.

6.4.1 | Sensitivity to deficit tolerance threshold, risk threshold, and duration

We set threshold levels for the parameters in Equation (6) to measure the sensitivity of any potential microgrid design. For z^*, we apply 0.0, 0.01, and 0.05 cutoffs for the highest acceptable deficit level in any given scenario; in other words, we track whether a potential microgrid design is able to satisfy the power loads all the time, at least 99% of the time, or at least 95% of the time in a given scenario. For ϵ, we apply 0, 0.01, 0.05, and 0.10 cutoffs for the highest acceptable risk tolerance, measured across all scenarios tested; in other words, we track whether a potential microgrid design meets the given z^* threshold in 100%, 99%, 95%, or 90% of the scenarios considered.

At the run level, there are $65 \times 180 \times 2 = 23\,400$ design points, each replicated 50 times for a total of 1.17 million simulated operations. 64.4% of these operations experience some power shortfall ($z > 0$). The deficit z ranges from 0.0 to 0.597 and is highly skewed, with a mean of 0.0406 and a median of 0.00119. This indicates that although operations often contain shortfalls, these shortfalls usually occur for only small amounts of time.

The impact of variability is striking as Table 2 shows. Of the 65 original rightsized designs, none met the most stringent $\{z *= 0, \epsilon = 0\}$ criteria. If we require all scenarios to meet power demand at least 99% of the time, only 10 of the original 65 microgrid designs are able to meet this performance

TABLE 2 Number of baseline rightsized designs satisfying deficit and risk threshold combinations.

		Two-week duration				Three-week duration			
		ϵ				ϵ			
		0	0.01	0.05	0.10	0	0.01	0.05	0.10
z^*	0	0	0	1	2	0	0	0	0
	0.01	10	22	28	31	9	17	22	26
	0.05	22	31	37	40	21	29	33	36

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standard. As we ease those requirements, either by allowing power deficits at more time steps or by allowing unacceptable deficits for more of the replications, more microgrid designs remain in consideration. In the most lenient case considered, where only 90% of scenarios must meet power demand only 95% of the time, 40 of the original 65 microgrid designs are able to meet this performance standard for a 2-week operation, and 36 are able to do so if the operation extends to 3 weeks. The small amount of excess capacity that arises because of the specified step sizes for component increments is often insufficient to absorb the variations in weather and power load we introduce.

For any $\{z^*, \epsilon\}$ pair, the viable microgrids for the three-week operation are a subset of those for the two-week operation. (In general, this may often but not always occur.) Note that the z^* and ϵ thresholds have different affects. When $z^* = 0$, few if any microgrids are acceptable even for large ϵ. The numbers of acceptable microgrids are much larger for large z^*, even when $\epsilon = 0$. We described some practical guidance for setting the thresholds in Section 3.4. From a methodological point of view, setting both of these thresholds to nonzero amounts—even if they are quite small—makes the results less sensitive to the size of the time step. Otherwise, order statistics tell us that the maximum observed deficit z tends to increase as the number of time steps increases. Similarly, as the number of replications increases, the probability that $Risk > \epsilon$ for some fixed $\epsilon > 0$ will stabilize, while the probability that $Risk > 0$ will continue to rise. The compositions of the microgrids that remain viable is also interesting; we will examine this further in Section 6.5.

6.4.2 | Sensitivity to demand profile and weather distributional parameters

The results in Section 6.4.1 reveal the uncertainty introduced by varying the demand and weather distributional parameters, but do not indicate whether these parameters are equally impactful. Fortunately, we can measure this sensitivity and then construct response surface metamodels to determine whether any portion can be attributed to the distributional factors we varied. Partition trees (also known as decision trees or classification and regression trees) and multiple regression are two useful types of metamodels (Ryan, 2007; Witten et al., 2017) for accomplishing this task.

Briefly, partition trees begin with all response data in a single group. The mean and standard deviation of the response are computed, and the initial explanatory power $R^2 = 0$. This large group is split into two smaller "leaves" of the tree using a simple rule or binning process for a single factor. For a quantitative factor, this split can occur anywhere between its minimum and maximum. For a categorical factor, the split bins the levels into two separate groups—one for each leaf. The optimal split is the one that leads to the greatest improvement in R^2. The splitting process continues, successively identifying the best leaf, factor, and split. The partition tree metamodel predicts the mean and standard deviation of the response using the values for its associated leaf. Multiple regression metamodels characterize the average response as a mathematical function of the factors, and also characterize the unexplainable variability or random error. In either case, the goal is to see whether a parsimonious metamodel can be constructed with a statistically significant and practically useful R^2 value.

For illustrative purposes, we consider two responses at the microgrid × scenario level, where there are $65 \times 180 \times 2 = 23\,400$ design points (from summarizing the raw results over 50 replications). This level of granularity is suitable for metamodels to explore the impact of the distributional parameters. The overall deficits and risks would be of interest for assessing the robustness of the microgrids, but not for attributing the robustness to different sources of variability.

Constructing both types of metamodels is informative. A partition tree may provide a better fit if there are jumps (discontinuities) or large flat regions in the response surface. If the true response surface is smooth and well behaved, a regression metamodel may capture its behavior more parsimoniously than a partition tree. If both types of metamodels yield similar insights, the choice of metamodel may depend in part on the audience or intended users of the study.

The first response of interest is mean $\left(z_j^{\omega}\right)$, the average deficit for microgrid j in scenario ω, which is the average proportion of time during a single operation where power needs are not fully met. The second is $Risk_{j,0}^{\omega}$, which is the risk of having any deficit for microgrid j in scenario ω.

The first five splits of a nine-split partition tree metamodel for mean (z) appear in Figure 7. After nine splits, $R^2 = 0.136$, up from $R^2 = 0.107$ after only five splits. This is low, indicating that a great deal of variation is not attributable to the distributional parameters. The leaves on the tree show, unsurprisingly, that lower values of z occur when the weather conditions are favorable—specifically, if the upper cloud limit c is high, the cloud ratio b is high, and solar scaling parameter μ is high. There is a slight increase in the mean (z) when the duration extends to three weeks. The column contributions shown in the bottom right of this plot indicate that c and b are the most important terms, together accounting for 89% of the meager 13.6% of its variability. Neither the autoregressive parameter ϕ nor the solar scaling parameter σ have any significant impact.

Stepwise regression yields similar results. It does not include ϕ or σ in any metamodel, and the four most important terms are also in the same: c, b, μ, and duration. A main-effects metamodel yields $R^2 = 0.154$, while a metamodel including four two-way interactions and three quadratic effects yields $R^2 = 0.177$.

The same four factors are important in the metamodels for $Risk_{j,0}^{\omega}$, although this risk is somewhat easier to predict. A partition tree with six splits yields $R^2 = 0.522$ and shows the most important factors are, in order, c, b, and μ as before.

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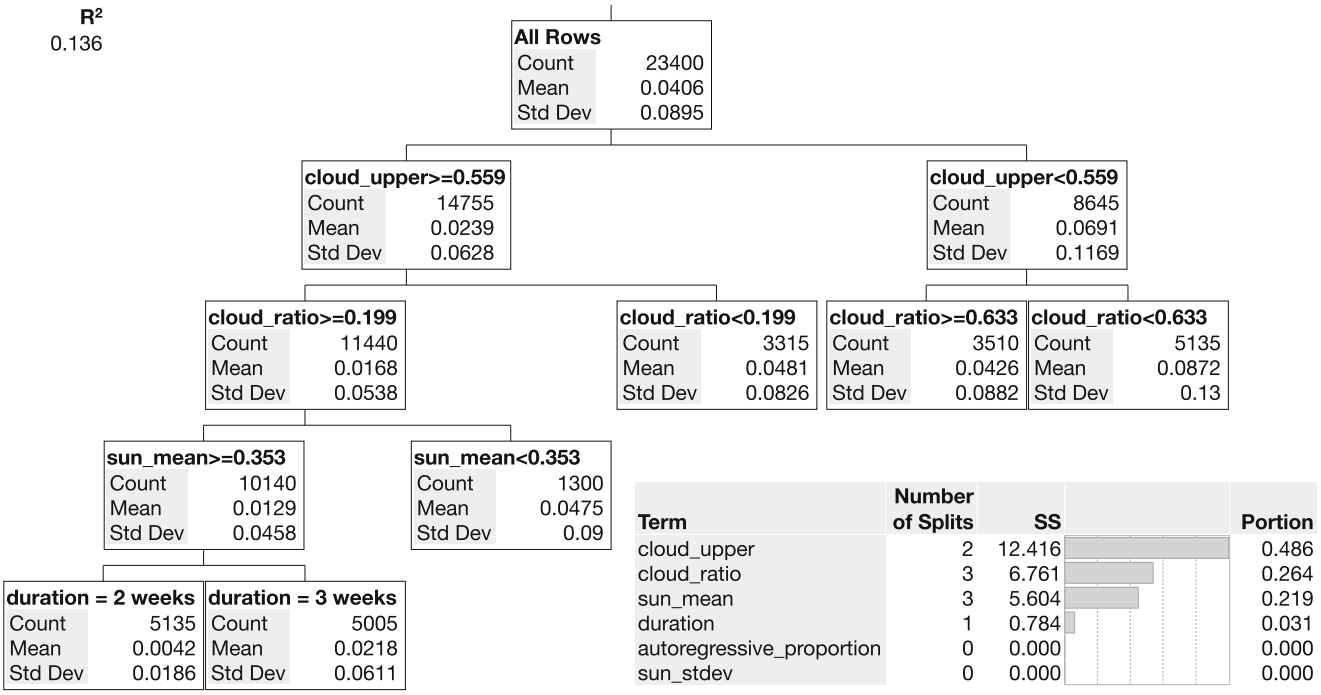

FIGURE 7 First five splits of a nine-split partition tree for the mean deficit z at the microgrid × scenario level.

Regression metamodels include the duration as the next most important term; we have $R^2 = 0.532$ for a model with four main effects, or $R^2 = 0.571$ for a model with four main effects, three quadratics, and four two-way interactions.

These findings have practical implications. They show that introducing variability means that many of the baseline rightsized microgrids will be unable to meet the specified demand profile in practice. Selecting microgrids that are more robust or resilient will make it more likely that operations will proceed smoothly from an energy perspective. From a modeling perspective, the results show that not all the distributional factors affect the response. From a data collection perspective, if the given model for ξ_t is appropriate, there is no need to collect detailed information to estimate the autocorrelation coefficient ϕ or the solar scaling parameter σ. This could potentially reduce the size of the design in subsequent experiments.

6.5 | Increasing capacity of rightsized microgrid designs

The set of potential microgrid designs that meet desired sensitivity threshold criteria may be insufficient. In such cases, we can consider alternative designs with increased capacity. In our experiments, we allowed capacity increases of 5%, 10%, 15%, and 20% to be applied to each of the 65 original potential microgrid designs, where these increases could be applied in increments of 5% to the photovoltaic and or battery energy storage systems. One of our 65 potential designs consisted of only diesel generators (120 kW capacity) and we did not consider expanding this design. We maintained the same threshold levels considered in Section 6.4.1 for the parameters in Equation (6). We restrict our presentation to 3-week operations in this section, but the results for 2-week operations are similar.

TABLE 3 Number of baseline rightsized microgrids that satisfy sensitivity thresholds after 20% capacity increases.

		Baseline original (65 microgrid designs)				Baseline expanded (901 microgrid designs)			
		ϵ				ϵ			
		0	0.01	0.05	0.10	0	0.01	0.05	0.10
z^*	0	2	3	5	10	2	4	9	20
	0.01	16	22	27	30	47	89	118	128
	0.05	22	31	36	38	92	136	160	173

The results for 20% increased capacity are shown in Table 3. In the most conservative case considered, where we require all scenarios to meet power demand all of the time, 2 of the original 65 microgrid designs are able to meet this performance standard, compared to none without the 20% capacity increase. In another conservative case where we require all scenarios to meet power demand at least 99% of the time, 16 of the original 65 microgrid designs are able to meet this performance standard, compared to 9 without the 20% capacity increase. As we ease those requirements, either by allowing power deficits at more time steps or by allowing more scenarios not to meet the power deficit threshold, the number of microgrid designs that remain in consideration increases. In the most lenient case considered, where only 90% of scenarios must meet power demand only 95% of the time, 38 of the original 65 microgrid designs are able to meet this performance standard, compared to 36 without the 20% capacity increase. We remark that the counts in the left portion of Table 3 correspond to the original 65 potential microgrids, not to the

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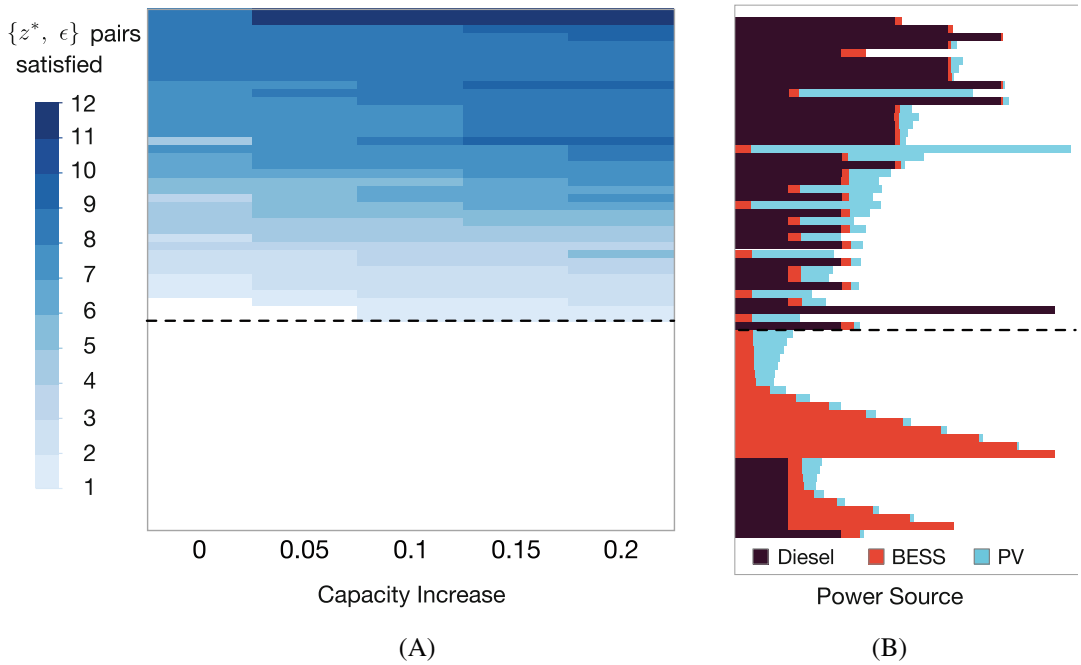

FIGURE 8 Consolidated view of microgrid designs and their ability to meet deficit and risk threshold combinations with or without capacity expansions. Those below the dashed lines fail to meet any combination. (A) Heatmap of occurrences, ordered from top (best) to bottom (worst) by the total count across all capacity increases, then by increasing diesel in the original design. (B) Microgrids with component-wise scaling, ordered from top to bottom as shown in (A).

modified capacity microgrids, because capacity increases can be distributed to photovoltaic and battery sources in a variety of ways that may meet the specified thresholds. The numbers of acceptable modified designs out of the total 901 modified designs that arise from the 65 baseline designs appear on the right portion of Table 3.

We now take a closer look at the compositions of the baseline rightsized microgrid designs and their expansions. In order to rank these grids, we consider the 12 $\{z^*, \epsilon\}$ pairs and count the number of times the microgrid satisfies a pair of criteria for each of five situations: expanded capacity ranging from 0% to 20% in increments of 5%. We determine an overall score (out of 60 possible), and sort the microgrids by this overall score (in decreasing order). In the case of ties, we sort in order of increasing diesel followed by increasing battery. The results appear in Figure 8.

Figure 8A is a heatmap of the occurrences, with the performance of the original rightsized designs on the left and the performances after expansion in subsequent segments. The darker the bar, the higher the count, so the microgrids that meet the most $\{z^*, \epsilon\}$ pairs are at the top. Figure 8B reorders the barchart of Figure 6B in the same way, so the composition of each microgrid is horizontally aligned with its row in the heatmap. Any microgrids appearing below the dashed lines on the subplots have overall scores of zero: even after a 20% expansion using photovoltaic or battery, they were unable to satisfy any of the risk criteria.

There are several insights we can draw from these plots. First, from Figure 8A, we see that the overall score rises slowly for many of the microgrids, as indicated by the fact that many of the rows contain just a few different shades. This shows that increasing capacity by 20% will not be of interest to all decision makers. Operations with very stringent requirements require microgrids or expanded microgrids that satisfy most or all of the $\{z^*, \epsilon\}$ pairs. However, unnecessarily large expansions could use up scarce resources or take up additional space, which might have detrimental affects on the operational budget or effectiveness without practical benefits in terms of acceptable risk.

Figure 8B shows that battery usage is typically much lower for the "good" microgrids (those above the dotted line) than for the "bad" ones below. Microgrids that rely heavily on battery sources of power are less robust to variations in the external environment, even when the decision makers accept some risks of power shortfalls.

6.6 | Discussion

The results for our computational example were intended to be illustrative, not exhaustive. Our framework can be applied using other model inputs—even a much larger number of factors. Instead of the $65 \times 257 \times 2 = 33\,410$ design points used to explore 65 microgrid designs at a fixed location and with a fixed start date for six distributional parameters at two durations as shown in Equation (11), we could have used a more efficient design as shown in Equation (10). For example, crossing the baseline rightsized grids with a single stack of an NOLH(11,33) requires just $65 \times 33 = 2145$ dps and allows us to add up to two other factors, while crossing it with a single stack using an NOLH(100 512) requires $65 \times 512 = 33\,280$ dps and allows us to add up to 91 other factors. Other designs are possible as well (Sanchez et al., 2021). New factors could

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include time-varying reliability of DER components, correlation between weather conditions and energy demand, embedded model constants such as the discharge efficiency or minimum allowable state of charge for BESS components, and more. Those interested primarily in long-term cost-effective solutions could include cost-related factors such as the availability of discounts, or trends in cost and improved efficiency over time. Those interested in long-term resilience to different types of vulnerabilities or threats could include the probability of various disruptions, together with distributional parameters describing their severity, for example, Li (2015) and Sanchez (2018). In short, the data farming approach is scalable and lends itself to much broader studies and different kinds of insights from those gained from either case studies or sensitivity analyses involving just a few factors.

One benefit of our framework is that not everything that can be modeled must be modeled. By providing sets of potential hybrid microgrid designs, rather than single alternatives, we provide greater flexibility to decision makers who may have considerations that are impractical or impossible to incorporate into an optimization or simulation model.

7 | CONCLUDING REMARKS

In this article, we develop a framework for producing a set of potential hybrid microgrid designs and assessing their robustness under a variety of conditions, including varying weather, varying power demands, and extended time horizons that exceed planning estimates. The ability to maintain operational viability in military applications was our motivating context for this work, but there are undoubtedly many other situations in which these considerations are essential. Our goal was to narrow down a set of options to those a decision-maker would consider and be able to weigh trade-offs. Accomplishing this goal required introducing an accessible notion of risk, attributable to power deficits simulated.

We pursued a decoupled approach, where the initial set of microgrid designs can be generated by any algorithm capable of doing so. We introduce capacity expansions for those microgrids under consideration, as a means of enhancing their operational capabilities. Our sensitivity analysis method is independent of the algorithms used to produce these microgrid designs, allowing it to be paired with other models currently available or ones developed in the future, for example, incorporating wind turbines, a DER not included in the model we use. By employing DOE in a data farming approach and parallelizing our implementation, we are able to maintain computational tractability, while considering a significant range of scenarios.

Finally, while our framework produces an abundance of information, we can succinctly present a set of viable options to a decision-maker. Risk can be considered alongside other important factors, such as fuel requirements, sustainability goals, weight, and cost to further narrow down the available options. We provide the relevant information and leave it to the decision-maker to select the most preferred microgrid design.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ORCID

Daniel Reich https://orcid.org/0000-0002-5605-5065
Susan M. Sanchez https://orcid.org/0000-0002-1976-3409

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